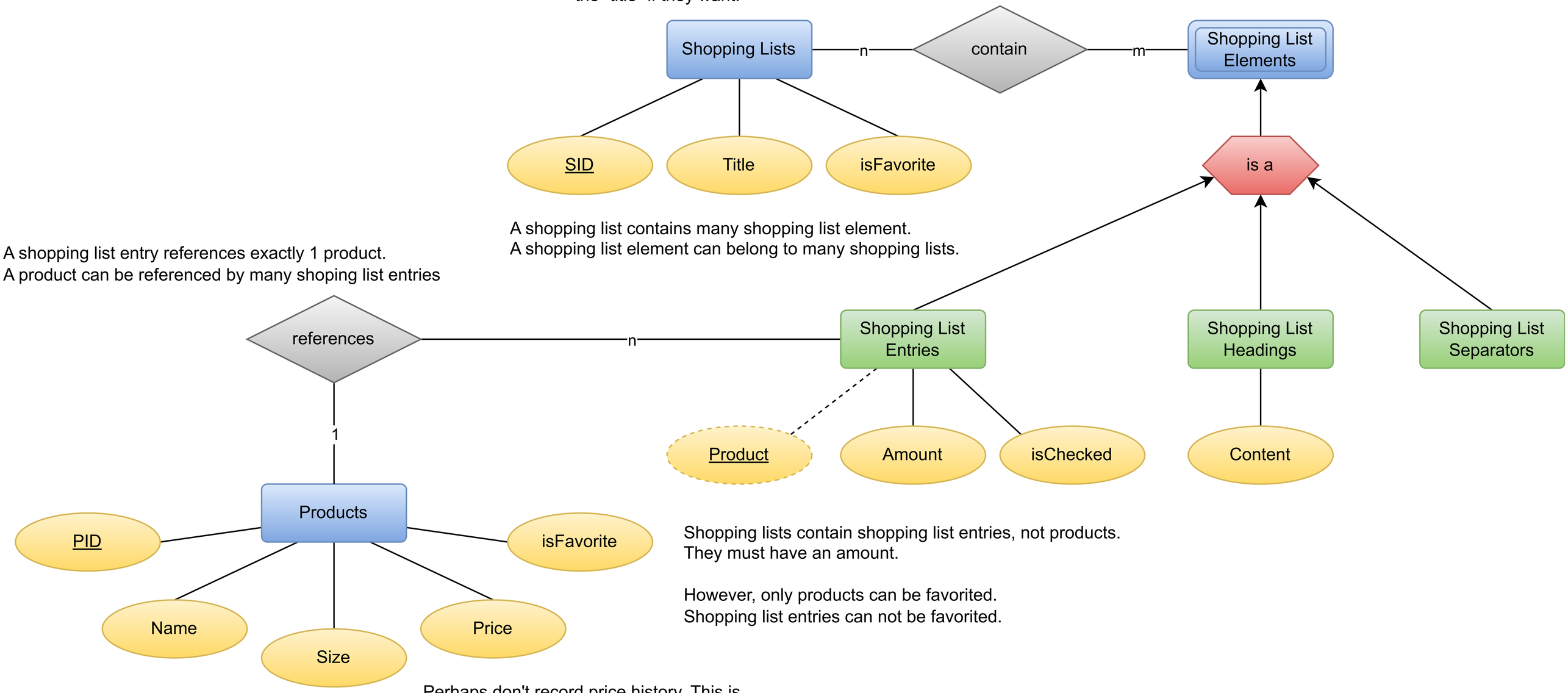
Should shopping lists have a date?
Perhaps leave it out, the user can put it in the `title` if they want.



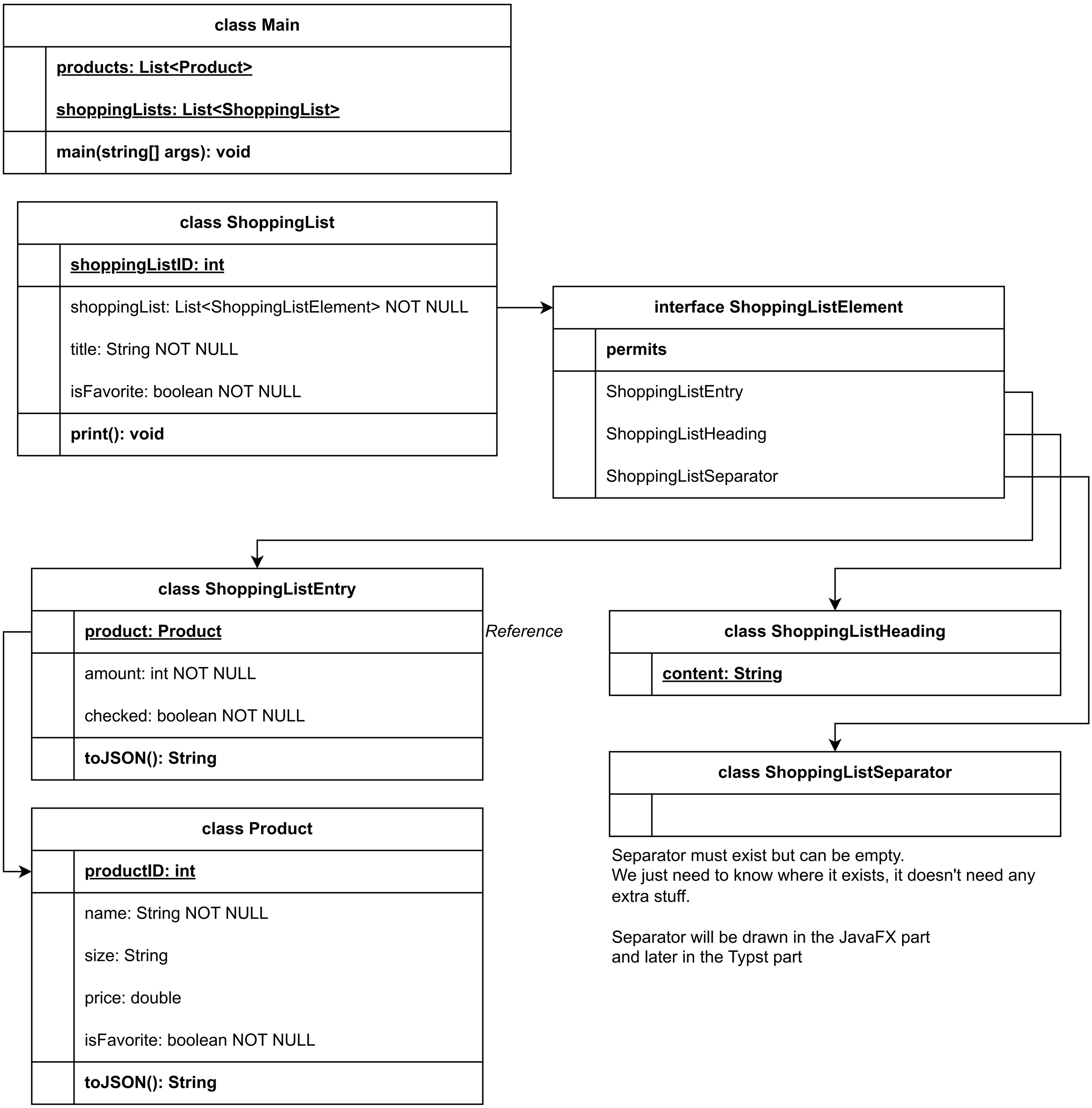
A shopping list contains many shopping list element.
A shopping list element can belong to many shopping lists.

A shopping list entry references exactly 1 product.
A product can be referenced by many shoping list entries

Shopping lists contain shopping list entries, not products.
They must have an amount.

However, only products can be favorited.
Shopping list entries can not be favorited.

Perhaps don't record price history. This is a simple shopping list manager; a price index can be a separate project


class Main
products: List<Product>
shoppingLists: List<ShoppingList>
main(string[] args): void
class ShoppingList
shoppingListID: int
shoppingList: List<ShoppingListElement> NOT NULL
title: String NOT NULL
isFavorite: boolean NOT NULL
print(): void
interface ShoppingListElement
permits
ShoppingListEntry
ShoppingListHeading
ShoppingListSeparator
class ShoppingListEntry
product: Product
amount: int NOT NULL
checked: boolean NOT NULL
toJSON(): String
Reference
class ShoppingListHeading
content: String
class ShoppingListSeparator
class Product
productID: int
name: String NOT NULL
size: String
price: double
isFavorite: boolean NOT NULL
toJSON(): String
Separator must exist but can be empty.
We just need to know where it exists, it doesn't need any extra stuff.
Separator will be drawn in the JavaFX part
and later in the Typst part

JSON format example

If we use JSON instead of something like SQLite, it could look like this:


```
{
  "products": [
    {
      "productID": 0,
      "name": "Milk",
      "size": "1L",
      "price": 1.99,
      "isFavorite": false
    },
    {
      "productID": 1,
      "name": "Eggs",
      "size": "10 pcs.",
      "price": 2.99,
      "isFavorite": false
    },
    {
      "productID": 2,
      "name": "Chips",
      "size": "250g",
      "price": 2.99,
      "isFavorite": true
    }
  ],
  "shoppingLists": [
    {
      "shoppingListID": 0,
      "title": "Untitled Shopping List",
      "isFavorite": false,
      "shoppingList": [
        { "type": "product", "productID": 0, "amount": 1 },
        { "type": "product", "productID": 1, "amount": 2 }
      ]
    },
    {
      "shoppingListID": 1,
      "title": "My Shopping List",
      "isFavorite": true,
      "shoppingList": [
        { "type": "heading", "content": "HOFER" },
        { "type": "product", "productID": 1, "amount": 1 },
        { "type": "separator" }
      ]
    }
  ]
}
```